

Mark Scheme (Results)

October 2024

Pearson Edexcel International Award
In Lower Secondary Computing (LCP11)
Paper 01

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

iLower Secondary Computing 2410

Section A

Question number	Answer	Mark
1a	Award **one** mark for: A URL is the web **address** that a user enters into a web browser.	**(1)**

Question number	Answer	Mark
1b	**The only correct answer is B** ***A*** *is not correct because GPS is a connectivity type* ***C*** *is not correct because a router directs data packets* ***D*** *is not correct because WAP is a protocol*	**(1)**

Question number	Answer	Mark
1c	Award **one** mark for any of the following: • So packets can be put back into the correct order (at the receiving end) • So packets can be sent in any order	**(1)**

Question number	Answer	Mark
2	Award **one** mark for any of the following: • Linked web pages • Linked content • **A collection of** web pages / websites / hypertext documents • Content stored on web servers Note Webs that can be viewed globally is just repeating the question and should not be awarded	**(1)**

Question number	Answer	Mark
3	**The only correct answer is B** ***A*** *is not correct because a hard disk drive is not read by a browser* ***C*** *is not correct because HTTP is a protocol* ***D*** *is not correct because HTTPS is a protocol*	**(1)**

Question number	Answer	Mark
4	Award **one** mark for: Connecting computers to the Internet allows people to **edit files together**	**(1)**

Question number	Answer	Mark
5	Award **one** mark for: A mobile phone network provides **wireless** connectivity.	**(1)**

Question number	Answer	Mark
6	Award **one** mark for any of the following: • Local (area network) / LAN • Wide (area network) / WAN • Virtual private (network) / VPN • Metropolitan (area network) / MAN	**(1)**

Question number	Answer	Mark
7	Award **one** mark for any of the following: • Bandwidth • Network (data) speed • (Data) transfer speed	**(1)**

Question number	Answer	Mark
8	**The only correct answer is A** ***B*** *is not correct because a microphone is an input device* ***C*** *is not correct because a microphone is an input device* ***D*** *is not correct because a microphone is an input device*	**(1)**

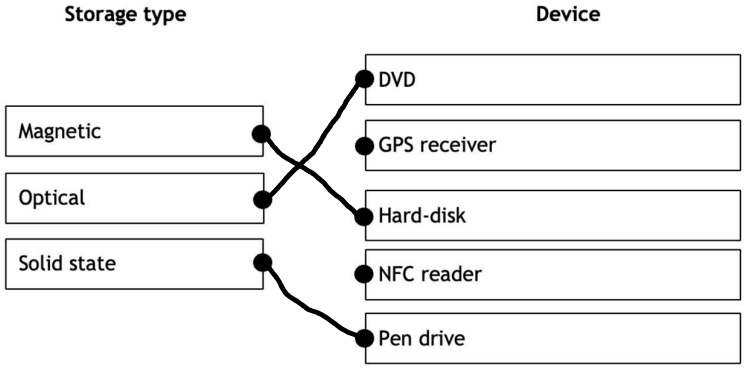
Question number	Answer	Mark
9	Award **one** mark for any of the following: • Open source • Free	**(1)**

Question number	Answer	Mark
10	Award **one** mark for: A user interface is provided by the **operating** system.	**(1)**

Question number	Answer	Mark
11a	Award **one** mark for: Data in random-access memory is lost when a computer is powered off.	(1)

Question number	Answer	Mark
11b	Award **one** mark for: Main memory stores **instructions** used by the central processing unit.	(1)

Question number	Answer	Mark
11c	**The only correct answer is B** ***A*** is not correct because anti-malware is not better if stored on a pen drive ***C*** is not correct because all storage should be backed up ***D*** is not correct because a pen drive uses solid state storage media	(1)

Question number	Answer	Mark
12	Award **one** mark for each correct line:  Do not award answers with more than one line to and from any of the boxes.	(3)

Question number	Answer	Mark
13a	Award **one** mark for: The code uses the value `"XL"` as an argument for the subprogram.	(1)

Question number	Answer	Mark
13b	**The only correct answer is C** ***A*** is not correct because `append` is the name of a method ***B*** is not correct because `del` is a keyword ***D*** is not correct because `XL` is a string	**(1)**

Question number	Answer	Mark
13c	**The only correct answer is A** ***B*** is not correct because S is the first item ***C*** is not correct because XL is the last item ***D*** is not correct because XXL is removed from the list	**(1)**

Question number	Answer	Mark
14a	Award **one** mark for: • 2	**(1)**

Question number	Answer	Mark
14b	Award up to **two** marks for a linked explanation such as: • There is a **syntax** error (1) because "count" is not a variable name (1) • There is a **syntax** error (1) because an integer cannot be assigned to a string (1) • "count" is not a variable name (1) because it is surrounded by quotes (1)	**(2)**

Question number	Answer	Mark
14c	**The only correct answer is D** ***A*** is not correct because a value is assigned to `letter` ***B*** is not correct because there is no iteration ***C*** is not correct because there is no selection	**(1)**

<table>
<tr><th>Question number</th><th>Answer</th><th>Mark</th></tr>
<tr><td>15</td><td>Award one mark for each correct cell:
<table><tr><th>Code</th><th>Output</th></tr><tr><td>4 % 2</td><td>0</td></tr><tr><td>3 // 2</td><td>1</td></tr></table></td><td>(2)</td></tr>
</table>

Question number	Answer	Mark
16	Question is two ways that digital technologies can harm the environment Award **one** mark for any of the following up to a maximum of **two** marks such as: • Burning fossil fuels / fires (caused by ewaste) • Unsafe disposal (of harmful components / ewaste) • Use of non-renewable materials (in manufacture) • Water pollution (caused by e-waste) • Damage to the landscape / environment eg deforestation, loss of farming land (caused by mining for materials used in manufacture • Air Pollution (caused by factory emissions) • Damage to human health/ animals / crops / plants (caused by factory emissions/exposure to toxic materials) Award answers that match cover the same ground as others in the list	**(2)**

Question number	Answer	Mark
17	Award **one** mark for any ways of responding to inappropriate conduct online up to a maximum of **two** marks: • Block • Report Answers may be in a sentence or using either of the two words	**(2)**

Question number	Answer	Mark
18a	Award **one** mark for: 1024	**(1)**

Question number	Answer	Mark
18b	**The only correct answer is C** ***A*** is not correct because the result of the addition is 1101 ***B*** is not correct because the result of the addition is 1101 ***D*** is not correct because the result of the addition is 1101	**(1)**

Question number	Answer	Mark
18c	Award **one** mark for: 8	**(1)**

Question number	Answer	Mark
18d	Award **one** mark for: Unicode	**(1)**

Question number	Answer	Mark
19ai	Award **one** mark for: 3	**(1)**

Question number	Answer	Mark
19aii	Award **one** mark for: 9	**(1)**

Question number	Answer	Mark
19b	Award **one** mark for: 3 Accept: 2	**(1)**

Question number	Answer	Mark
19c	Award up to **two** marks for a linked description such as: When there have been no swaps / all items are sorted (1) in a completed pass (1)	**(2)**

Question number	Answer	Mark
19d	Award **one** mark for: • linear ~~Although knowledge of these is not expected at this level, accept other search algorithms such as binary, DFS, BFS, Hashing, Interpolation, Exponential, middle number algorithm (not search)~~	**(1)**

Question number	Answer	Mark
20a	Award up to **two** marks for a description such as: • more values available (1) to represent the amplitude/sample (1) Answers rarely say how increasing bit depth improves the representation of sound • (higher bit depth) allows for more resolution/ greater variety of amplitude values (1) and improves the accuracy of recordings / details captured with greater precision / (1)	**(2)**

Question number	Answer	Mark
20b	A linked description that includes any **three** from: • A bitmap is represented using pixels • Pixels are arranged in a grid • Pixels have a number of colours options (colour depth) • Colour depth is determined by the number of bits used for each pixel • Resolution is affected by the number of pixels per inch ~~Although knowledge of these is not expected at this level, award reference to: 'metadata' (horizontal resolution, colour model, file format, compression).~~	**(3)**

Question number	Answer	Mark

21	Award **one** mark for each of:
	
	• `score`
	• `score` (or `10`)
	• `score` (or `10`) (check FT)
	• `,` or `+str(score)`
	
	Example:
	`3 score = score + 10`
	or
	`3 score = 10 + score`
	
	`4 print("Your score is: " , score)`
	or
	`4 print("Your score is: " + str(score))`
		(4)

Section B

Question number	Answer	Mark
22a	**The only correct answer is B** ***A*** is not correct because operators can be used in single tables ***C*** is not correct because queries can run on data in a single table ***D*** is not correct because records exist in single tables	**(1)**

Question number	Answer	Mark
22b	**The only correct answer is C** ***A*** is not correct because foreign keys are also used in offline databases ***B*** is not correct because foreign keys are used to create relationships ***D*** is not correct because a foreign key can have any name	**(1)**

Question number	Answer	Mark
22c	**The only correct answer is D** ***A*** is not correct because AVG is not a logical operator ***B*** is not correct because DIV is not a logical operator ***C*** is not correct because LIKE is not a logical operator	**(1)**

Question number	Answer	Mark
22d	Award **one** mark for: In a database, the **Boolean** data type has one of two possible values.	**(1)**

Question number	Answer	Mark
23	Award **one** mark for: Images can be edited by adjusting the **brightness** and contrast.	**(1)**

Question number	Answer	Mark
24	**The only correct answer is D** ***A*** is not correct because compression changes the file size, not image dimensions ***B*** is not correct because use of erase sets the colour of pixels to 'transparent' ***C*** is not correct because recolour changes the colour of pixels.	**(1)**

Question number	Answer	Mark
25	Award **one** mark for any of the following: • Makes the upload/download/**share** quicker • Faster transfer • Reduces the file size Do not accept: 'quicker/faster' without reference to upload/download/transfer/ **share**. Do not accept: 'smaller' without reference to file size.	**(1)**

Question number	Answer	Mark
26	Award **one** mark for any of the following: • It uses points / lines / **curves /** coordinates • It uses less file space • It does not hold data for the colour of each pixel • It is not as photo-realistic • It can be resized without loss of quality • scalable	**(1)**

Question number	Answer	Mark
27a	Award **one** mark for: In presentation software, embedding **multimedia** content, such as video, enhances the user experience.	**(1)**

Question number	Answer	Mark
27b	Award **one** mark for: A presentation aimed at young children should use **images** rather than text.	**(1)**

Question number	Answer	Mark
28	**The only correct answer is D** ***A*** is not correct because abstraction is not related to file access ***B*** is not correct because collaboration is what is possible once permissions have been set ***C*** is not correct because formatting a disk erases files	**(1)**

Question number	Answer	Mark
29	**The only correct answer is C** ***A*** is not correct because disk utilities do not provide access to cloud-based software.	**(1)**

	B is not correct because file handing is not a cloud-based service ***D*** is not correct because a web filter does not provide access to software.	

Question number	Answer	Mark
30	Award **one** mark for any of the following: • To control the layout • To start a new page	**(1)**

Question number	Answer	Mark
31a	Award **one** mark for any of the following up to a maximum of **two** marks: • Bold / font enhancement • Alignment • Paragraph spacing	**(2)**

Question number	Answer	Mark
31b	Award **one** mark for any of the following: • A memo is a shorter communication • A letter is more formal • A memo is used internally Accept alternatives for letter or memo e.g. a memo is less formal	**(1)**

Question number	Answer	Mark
32a	Award **one** mark for: percentage	**(1)**

Question number	Answer	Mark
32b	Award **one** mark for any of the following: • Pie • Bar	**(1)**

Question number	Answer	Mark
32c	Award **one** mark for each of: =B8 (1) / (1) B1 (1) Correct answer: =B8/B1	**(3)**

Question number	Answer	Mark
32d	Award up to **two** marks for a linked explanation such as: • Data is clearer to read / easier to find (1) because it is separated / because a summary sheet can be produced (1) • Data can be formatted differently (1) such as different column widths (1)	**(2)**

Question number	Answer	Mark
33	Award up to **two** marks for a linked description such as: • A set of consistent design conventions (1) used by an organisation (1) • A set of consistent design conventions (1) to present documents and communications in a recognisable manner (1)	**(2)**

Question number	Answer	Mark
34	Award up to **two** marks for a linked explanation such as: • Information is more reliable (1) because more viewpoints can be seen/analysed (1)	**(2)**

Question number	Answer	Mark
35a	Award **one** mark for: bookmark	**(1)**

Question number	Answer	Mark
35b	Award up to **two** marks for a linked description such as: • Elements could be styled (1) using HTML tags / CSS (1) • Style tags (1) by adding element property values (1) • Add images (1) using the <img> tag (1) • Use <h1> tags (1) to style headings (1) • Style <p> (1) to format paragraphs (1) • Add tags (1) to break up blocks of text (1)	**(2)**

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